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Project Title:	Eutectic Solidification in a Ternary Alloy		
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Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
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Space Biology Element:	None		
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Zip Code:	35294-4461	**Congressional District:**	7
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/14/2021 per NSSC info (Ed., 8/26/20) NOTE: Extended to 8/14/2020 per NSSC info (Ed., 7/31/19)		
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Task Description:	Note this is a follow-on project for Dr. Ralph Napolitano's project,"Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy," which ended in 2014. Solidification processing of metallic alloys can produce a wide array of growth morphologies, which can be manipulated to access new structures, materials, and functionalities. The array of accessible structures is particularly wide in ternary and other higher-order systems, but the challenge of controlling the many variables that influence growth structures is immense and requires a better understanding of the interplay between the mechanisms of pattern formation and the influence of applied forces and processing conditions. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal field. The two principle objectives to be addressed are (i) to establish an analytical framework for 2D and 3D quantitative description of multiphase microstructures, and (ii) to employ the newly developed quantitative framework to investigate the dynamics of pattern formation in ternary eutectic alloys. The work will focus primarily on the invariant three-phase eutectic structure. Ground-based samples will be processed via a vertical Bridgman-style directional solidification furnace. Electron backscatter diffraction will be a major characterization tool to measure crystallographic orientation, texture, and misorientation relationships. Analysis over the length of the sample will provide insight into initial transient behavior, grain competition, and steady state growth. The relationship between geometric and crystallographic parameters, such as curvatures, orientations, and periodic ordering, will be used to quantify the morphological trajectory of the eutectic and identify fundamental principles of pattern formation. The project operates in collaboration with the European Space Agency’s project titled “Solidification along a Eutectic Path in Ternary Alloys” (SETA), which has already accomplished important work using the Al-Ag-Cu alloy system. Significant findings of the SETA project to date have been in the areas of pattern development and the role of fault structure for both univariant and invariant eutectic compositions, the role of crystal orientation and “locked” grains, and development of nonplanar growth modes. Facilitating open scientific interaction between the Principal Investigator (PI) and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments using the Materials Science Lab (MSL) and the Low Gradient Furnace (LGF), intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. The planning and performance of flight-based solidification experiments is being done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth. Additional flight samples through this project were originally planned but did not come to fruition.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled, multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is far-reaching, given the broad range of multiphase structures that can be fabricated through solidification processing, and is not limited to metals and alloys. Indeed, the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
	The major accomplishments of the project are summarized below. The technical activities during the final year focused on (i) analysis of the effect of velocity and thermal gradient on complex eutectic growth in the Al-Cu-Mg system as a comparison to previous work on Al-Cu-Ag, and (ii) evaluation of data obtained by tomography, with particular emphasis on grain competition. Planned work using principal component analysis to better quantify ternary eutectic microstructures and their response to imposed conditions and on modeling solid state diffusion in Al-Cu-Ag ternary eutectic growth did not come to fruition due to lack of progress by collaborators. EQUIPMENT DEVELOPMENT: During Year 1 of the project, a Bridgman-type furnace for directional solidification of samples was designed and built to the PI’s specifications by a team of undergraduate students as their senior design project. A thermal gradient is created inside the furnace chamber by a Kanthal wire resistance heating element supported by a mullite shell and a water-cooled copper chill block. The heating element was selected specifically to support the range of temperatures needed to investigate Al-Ag-Cu (700-1200°C). The chamber is insulted with two types of zirconia insulation. A screw drive and stepper motor control vertical translation of the furnace unit. Samples are contained in a suspended alumina crucible. A quick release mechanism was designed to allow the crucible and sample to drop smoothly into a container of water to accomplish quenching. The furnace sits on a levitated, vibration damping table to minimize external influences on microstructure development during processing, and an inert argon atmosphere is maintained by direct attachment of a tube to the crucible. This furnace forms a key component of the PI’s laboratory today and continues to be used extensively. DEVELOPMENT OF IMAGE ANALYSIS TOOLS: A major component of this project was to identify meaningful parameters for quantifying complex three-phase eutectics, which unlike binary eutectics cannot be fully characterized by a single eutectic spacing value. Before parameters can be measured, grayscale images from a microscope must be thresholded and segmented, which is also a far more challenging endeavor than for two-phase images, as the computer tends to mistake black/white phase boundaries as regions of the gray phase. MatLab code was developed and tested by the group which can accurately threshold a variety of three-phase eutectic structures in a semi-automated way. The final version of the code uses a multi-threshold image segmentation algorithm based on Otsu's method. Additional MatLab code was developed to carry out a variety of types of quantifications of the microstructures, including volume fraction, area and perimeter of each phase region, specific surface, shape factor, major and minor axis length, and distances. Subsequent iterations of the code added measurement of the length of the interfaces between the particles, the angular orientation of these interfaces, the lamellar spacings, and the number of nearest neighbors. Later in the project, more ambitious methods of characterizing and quantifying the microstructures were developed, based partly on different methods of visualizing the structures. Other techniques include measuring and coloring phase regions to identify changes in boundary curvature, maximum curvature, shape factor, sphericity and orientation of major axis. This last measure proved particularly useful, as the usual chain-like structure of the Al-Ag-Cu ternary eutectic with

Task Progress:

elongated intermetallic “links” means that the axis orientation of either the Al2Cu or Ag2Al phase can be used as a marker for chain alignment and therefore grains. All thresholding and analysis code developed through this program can be used in the future for analysis of a variety of complex microstructures.

ACQUISITION OF 3D TOMOGRAPHY DATA: In order to more completely capture the complex evolution of the ternary eutectic structure, tomographic data was collected at the Advanced Photon Source at Argonne National Lab from samples processed at University of Alabama at Birmingham (UAB). Two regions (diameter = 0.5 mm, length = 6-7 mm) were machined from a variety of samples processed at different constant velocities, samples with different velocity changes, and samples extracted from both regions of initial solidification and the quenched interface so that the microstructure evolution in these transient regions could also be studied. This data is in addition to tomography data of the same alloy obtained by SETA members of ground-based samples in 2015 and flight samples in 2020.

Using the method developed in Year 6 to segment the relatively low resolution and low contrast tomography images, along with the orientation coloring technique on Ag2Al phases described above, the movement of grain boundaries and individual rods with different grains could be tracked. Tracking x:y coordinates with increasing growth height allowed for the tilt angle of the microstructure of each grain with respect to the direction of heat extraction to be calculated. Using the radial plots allows both the growth direction and magnitude of tilt to be displayed and directly compared. The final analysis of this data for publication is still being completed. The tracking of the rods also proved to be a useful way to find locations where rods merged or split, providing new experimental data for comparison with previous simulation work published by the group and European partners on this topic.

IDENTIFICATION OF PRIMARY ORIENTATION RELATIONSHIPS (OR) IN AL-AG-CU EUTECTIC: Although crystallographic orientation analysis of invariant ternary Al-Ag-Cu eutectic had been undertaken by various SETA partners since at least 2009, very little about the crystallographic orientation relationships in this system had been published. In Years 4 and 5, the group finally brought together all electron backscatter diffraction (EBSD) data on this system collected by us and various members of the SETA project to conduct a systematic review and analysis. The result was the 2018 publication in Acta Materialia of the primary/preferred orientation relationship for the system, which is based on maximization of low energy {110} planes in Al2Cu, along with other commonly observed ORs. Possible mechanisms for adjustment of eutectic spacing as a result of changing solidification conditions were also proposed and illustrated in this paper.

SOLID STATE MICROSTRUCTURE EVOLUTION: One challenge in studying the ternary eutectic system Al-Ag-Cu is a relatively large solubility shift of the chemical element Ag in the Al phase after solidification, which influences both the phase fractions and composition. It is necessary to clearly understand the effect of these solid-state changes in order to separate them from the original solidification behavior. This is particularly important for the evaluation of flight samples, since those samples must be processed without quenching and only microstructure influenced by solid state diffusion is therefore available for analysis.

One way to observe solid state effects is to quench the sample and observe the microstructure at different distances behind the quenched interface which was done by the previous PI from Iowa State and published together with the current PI during Year 1 of this project. Based on these results, a cellular automaton model was developed to evolve the microstructure forward or backwards in position. Application of the model produced a reasonable estimation of the structure at the growth front based on the structure measured at some known distance behind the front.

While the quenching technique allows observation of microstructure evolution within a particular grain, it does not permit a specific region of microstructure to be observed at various stages. To overcome this limitation, directionally solidified samples were quenched and systematically annealed just below the eutectic temperature. This mimics the process that occurs naturally during cooling in the directional solidification furnace. The results showed that the phase fraction changes in the micrographs are in good accordance with the theoretical prediction from the thermodynamic Calphad database. Different patterns exhibited different behavior during annealing, resulting in a different high-temperature stability. For the chain-like structure, the pattern is preserved even after annealing for 32 h at more than 96% of the melting temperature, which underlines the capability of eutectics for high temperature applications. The interfaces become straighter and more aligned, which demonstrates the influence of the crystallographic anisotropy. For more irregular arrangements the pattern is not preserved; instead the rods merge together and the interfaces become rounder. It was also seen that phase fraction adjusts much more quickly than shape factor, which could be an important factor in deciding which parameter(s) to use to characterize and compare different microstructures.

This work was published in 2019 and the data, including images of additional grains not included in the paper, were provided to collaborator Prof. Britta Nestler at Karlsruhe Institute of Technology (KIT), to provide benchmarking data for phase field code they are currently developing that will incorporate solid state diffusion into the formation of ternary eutectic microstructures for the first time.

COMPARISONS OF EXPERIMENTAL AND SIMULATED EUTECTIC GROWTH: Throughout the duration of the project, the UAB team has enjoyed a highly fruitful collaboration with the phase field modeling group at KIT, resulting in five joint publications. Excellent agreement could be achieved between simulated and experimental microstructures. The simulated microstructure assumed the chain-like pattern typical for Al-Ag-Cu, even though the code used fully isotropic solid-solid interfacial energies. This indicates that there are diffusional drivers for the pattern as well as the crystallographic ones discussed above.

One major topic of investigation with phase field modeling was the response of the eutectic to imposed velocity changes and the role of splitting, merging, termination, and possibly nucleation events as mechanisms for spacing adjustment. In addition to comparisons of simulation with 2D cross sections, tomography data was also included in the analysis. One particularly unexpected finding was that the phase fractions were found to change slightly with velocity, both in the simulations and the experimental data. Until now, no explanation has been found, but further analysis of flight samples may provide additional insight. In addition, it was found that investigated statistics respond to velocity changes at different rates. The phase fractions adjust immediately after the velocity changes, while the reaction of the microstructural pattern occurs more slowly, as measured by shape factors and nearest neighbor statistics. It is a particularly significant result that the analysis show that these structures continue to evolve as they freeze, moving very slowly toward a more well aligned structure, but very slowly. This emphasizes the importance of long duration experiments such as those done on the ISS.

EUTECTIC GRAIN COMPETITION: Another major theme of research that has been present over the course of the project is the topic of grain competition: how regions with different patterns and/or orientation relationships increased or

decreased in relative size as growth progressed. This issue has been considered using image analysis with and without EBSD data, from sequential 2D sections at different growth heights and 3D tomography data. Observations of interactions at grain boundaries showed for the first time that higher order eutectic grain boundaries are not closed; that is, phase regions from one grain sometimes grow into another grain, forming mixed subgrains. Additional insights into the preferred crystallographic orientations were gained by considering not just random grains from a variety of samples, but a selection of grains from different growth heights of one or two samples.

COMPARISON OF AL-CU-AG TERNARY EUTECTIC WITH AL-CU-MG: The European SETA project considered only eutectics within the Al-Cu-Ag alloy system. To compliment this work, the research team at UAB also undertook analysis of one of the ternary eutectics in the Al-Cu-Mg system. This eutectic shares two phases with that of Al-Ag-Cu: solid solution Al with its face-centered-cubic crystal structure and Al2Cu, which is tetragonal. However, the difference of the third phase (hexagonal Ag2Al vs orthorhombic Al2CuMg) makes for an interesting comparison between the two systems. A systematic investigation of the effects of both solidification velocity and thermal gradient on the microstructure formation and evolution was undertaken. Processing at the highest gradient was carried out by SETA partner Access e.V. in Germany, as well as some of the EBSD analysis.

Samples of the invariant eutectic composition were solidified with varied thermal gradients (G = 3 - 14.4 K/mm) and velocities (0.625 - 4 µm/s). The results of the qualitative and quantitative assessments of the evolving structure showed that the modified two-lamellar, one-rod microstructure previously presented for this system breaks down below a critical velocity, with the a-Al and Al2CuMg phases losing their lamellar character and the three phases assuming a more equiaxed configuration. The system was found to maintain stable, three-phase coupled growth over a range of liquid compositions, solidification velocities and thermal gradients, and to be able to accommodate a range of compositions within the ternary eutectic itself. By adjusting processing conditions, pure ternary eutectic or a bimodal structure consisting of two-phase cells and highly regular ternary eutectic could be produced. By increasing the thermal gradient, the stability of the solid/liquid interface was changed, to produce cellular-dendritic, cellular, and planar growth.

EBSD analysis showed that crystal growth directions in all grains observed were found to match those reported in previous literature. However, the growth directions did not always align perfectly with each other or with the z-axis (solidification direction). Angular measurements of misorientation suggest that a very strong driving force for this particular growth alignment exists in the system, where any regions with markedly different alignments are almost immediately overgrown. This is unlike the Al- Ag-Cu ternary eutectic system where highly tilted crystals (particularly of Ag2Al) have been observed to persist much longer. However, the alignment to the true growth direction was noticeably better in the grains measured at higher positions in the sample, indicating that a grain competition process continues to operate in the Al-Mg-Cu samples over much longer distances, as those grains which are most optimally aligned with the direction of heat extraction slowly overgrow those which are slightly tilted.

The OR at the lamella-lamella boundary was also found to be the same as that reported in the literature, which corresponds to an interface with a high degree of lattice matching. However, at very high gradients, some grains were found with wavy lamellar interfaces, meaning that the system assumes the lamellar/lamellar/ rod phase arrangement even when that low energy boundary does not appear in the microstructure. EBSD measurements could not be made on these grains, leaving an intriguing open question. Close examination of the microstructure revealed an unexpected modification of lamellar growth, with lamellar sheets being limited in length to approximately four times the width of one sheet, and the Al2Cu rod phase aligning along these pre-existing boundaries which correspond to low index crystallographic planes. Alignment of the pattern was found to increase with increasing thermal gradient. The high gradient structure begins to resemble the chain-like pattern observed in Al-Ag-Cu eutectic, except that the “lamellar” phase (in this case, Al2Cu rather than Al as in the Al-Ag-Cu system) is periodically interrupted. This arrangement has been shown to be preferred in the Al-Ag-Cu system for both energetic interfacial reasons and diffusional reasons, forming in directionally solidified phase field simulations even when all interfaces are isotropic. Whether similar reasons are responsible for the pattern’s appearance at high thermal gradients in Al-Cu-Mg eutectic is an interesting question for future research.

Bibliography Type:	Description: (Last Updated: 06/05/2023)
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